

Generating Alpha: A Hybrid AI-Driven Trading System Integrating Technical Analysis, Machine Learning and Financial Sentiment for Regime-Adaptive Equity Strategies

Data:

1. Historical Price Data

- **Type:** Structured / numerical (time-series)
- **Source:** yfinance API
- **Specific data:** OHLCV — Open, High, Low, Close, Volume
- **Coverage:** 100 large-cap S&P 500 tickers across technology, financials, energy, healthcare, and consumer discretionary sectors
- **Time span:** 01/01/2019 to 01/01/2025 (6 years)

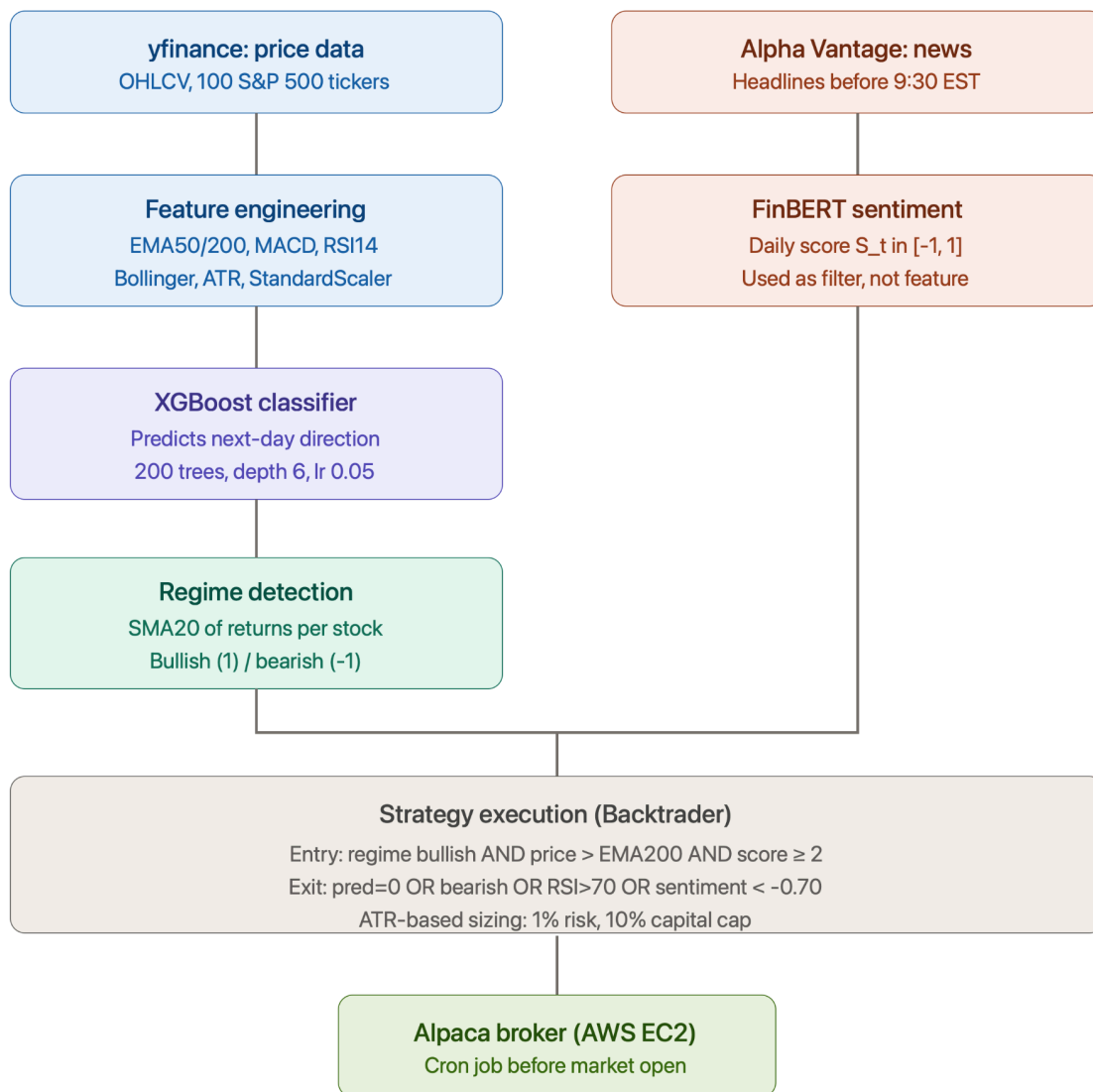
2. Financial News Data

- **Type:** Unstructured / textual
- **Source:** Alpha Vantage News Sentiment API (paper also loosely mentions FinancialModelingPrep)
- **Specific data:** News headlines and summaries, per ticker per date
- **Constraint:** Only articles published before 9:30 AM EST on the trading day (to avoid look-ahead bias)

Derived data (computed from the price data above, not separately collected):

- **Technical indicators** (structured/numerical, via TA-Lib): EMA-50, EMA-200, MACD + signal line, RSI(14), Bollinger Bands, ATR, rolling volatility
- **Interaction features:** EMA50/EMA200 ratio, MACD histogram (MACD – signal)
- **Sentiment scores** (numerical, derived from news text via FinBERT): normalized daily per-ticker score in range $[-1, 1]$
- **Market regime labels** (categorical: bullish/bearish, from 20-day rolling average return)

We have to include some slippage and transection cost on top of the results.



The data are mostly free here.